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A Related Work

Multimodal Math Reasoning The mathematical reasoning capabilities of MLLMs have recently attracted significant attention [11, 80, 35, 81, 82, 5, 14, 78]. Unlike traditional mathematical reasoning tasks in Language Models (LLMs) [1, 83], multimodal mathematical reasoning requires MLLMs to interpret visual information and perform cross-modal reasoning between images and text. Tasks such as solving geometric problems and analyzing graphs are particularly challenging [40]. Recent advances have focused on improving visual mathematical input specialized encoders in specific scenarios [19, 84, 61]. A significant emphasis has also been placed on synthesizing diverse and complex training data. For instance, Math-LLaVA [15] introduces the MathV360K dataset, which categorizes images by complexity and enhances associated questions. Multimath [23] curates high-quality reasoning data from K-12 textbooks and employs GPT-4 for CoT data generation and validation. R-CoT [17] further diversifies problems through a two-stage reverse question-answer generation process. These data synthesis methods are widely adopted in academia and industry due to their demonstrated efficiency [85–88].

Process Reward Model Recent studies have explored test-time scaling laws in LLMs, aiming to identify optimal reasoning ecectories from diverse thinking trajectories [26, 89–91, 31, 92, 93]. Initial efforts, such as self-consistency [79], have laid the groundwork for test-time scaling. OpenAI has introduced verifiers to supervise and select reasoning paths during inference [41]. Math-Shepherd [43] evaluates intermediate reasoning steps based on their likelihood of leading to correct answers, while OmegaPRM [42] constructs PRM training data and employs MCTS for training. Despite these advancements, the lack of models with robust CoT reasoning capabilities and limited exploration into diverse reward model training data remain significant bottlenecks in multimodal mathematical reasoning. Some concurrent work also begins to pay attention to PRM-assisted visual reasoning, such as construction and benchmarking [94–96].

B Preliminary

B.1 Group Relative Policy Optimization

Vanilla GRPO eliminates value function in PPO and estimates the advantages within online rollout group. Given a question with image q and ground-truth y, policy model $\pi_{\theta_{old}}$ samples a group of G responses $\{o^i\}_{i=1}^G$. GRPO compute the i-th response's advantage through normalizing in-group rewards $\{r^j\}_{j=1}^G$, and employs PPO's clipped objective and KL penalty term:

$$A^i = \frac{r^i - \text{mean}(\{r^j\}_{j=1}^G)}{\text{std}(\{r^j\}_{j=1}^G)} \tag{7}$$

$$\begin{aligned}\mathcal{J}_{GRPO}(\theta) = \mathbb{E}_{(q,y)\sim\mathcal{D},\{o^i\}_{i=1}^G\sim\pi_{\theta_{old}}(\cdot|q)} \\ [\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o^i|}\sum_{t=1}^{|o^i|}(\min(r_t^i(\theta)A^i, \text{clip}(r_t^i(\theta), 1-\epsilon, 1+\epsilon)A^i) - \beta D_{KL}^{i,t}(\pi_\theta||\pi_{ref}))]\end{aligned} \tag{8}$$

We introduce the two variants of PRM-integrated GRPO discussed in Section 4. Given PRM $\mathcal{M}_p$ and process reward sequences $r_s = \mathcal{M}_p(\{s_1, s_2, \cdots, s_N\})$ (i) *Variant 1*: Given verifiable outcome reward r_o^i, we set a single rollout's reward as $r^i = r_o^i + \bar{r_s^i}$. (ii) *Variant 2*: We utilize a step-level reward and a multiple relative advantage calculated by the mean value of process rewards from each rollout:

$$A_t^i = \underbrace{r_{s,t}^i \frac{\bar{r_s^i} - \text{mean}(\{\bar{r_s^j}\}_{j=1}^G)}{\text{std}(\{\bar{r_s^j}\}_{j=1}^G)}}_{\text{GRPO with process rewards}} + \underbrace{\frac{r_o^i - \text{mean}(\{r_o^j\}_{j=1}^G)}{\text{std}(\{r_o^j\}_{j=1}^G)}}_{\text{GRPO with outcome rewards}} \tag{9}$$

in which $\bar{r_s^i} = \text{mean}(\mathcal{M}_p(\{s_1^i, s_2^i, \cdots, s_{T_i}^i\}))$.